Teorema Master: $T(n) = a \cdot T\left(\frac{n}{b}\right) + f(n)$; $a \geq 1; b > 1$

Caz I: daca $f(n) = \theta(n^c)$ unde $c < \log_b a$

$\Rightarrow T(n) = \theta\left(n^{\log_b a}\right)$

Caz II: daca $f(n) = \theta\left(n^c \cdot log^k n\right)$ pentru $k \geq 0$ unde $c = \log_b a$

$\Rightarrow T(n) = \theta\left(n^{\log_b a} \cdot log^{k+1} n\right)$

Caz III: daca $f(n) = \theta(n^c)$ unde $c > \log_b a$

$\Rightarrow T(n) = \theta(f(n))$

Exercitiul 1

$T(n) = 4T\left(\frac{n}{4}\right) + 1$

$\Rightarrow a = 4;\ b = 4;\ c = 0;\ f(n) = 1 \qquad \log_b a = \log_4 4 = 1 > c$

$\Rightarrow Master\ I \Rightarrow T(n) = \theta\left(n^{\log_4 4}\right) = \theta(n)$

$T(n) = 8T\left(\frac{n}{2}\right) + 2n^2$

$\Rightarrow a = 8;\ b = 2 \quad \log_b a = \log_2 8 = 3 \quad f(n) = 2n^2 \in \theta(n^2) \Rightarrow c = 2 < \log_b a$

$\Rightarrow Master\ I \Rightarrow T(n) = \theta\left(n^{\log_b a}\right) = \theta(n^3)$

$T(n) = 4T\left(\frac{n}{2}\right) + n^2$

$\Rightarrow a = 4;\ b = 2 \quad \log_b a = \log_2 4 = 2 \quad f(n) = n^2 = \theta(n^2 \cdot log^0 n)$

$\Rightarrow c = 2 = \log_b a\ ;\ k = 0 \Rightarrow Master\ II \Rightarrow T(n) = \theta\left(n^{\log_b a} \cdot log^{k+1} n\right) = \theta(n^2 \log n)$

Exercitiul 2 – Binary Search

```
int search(int arr[], int lo, int hi, int v)
{
    int mid = (lo + hi) / 2;
    if (arr[mid] == v)
        return mid;
    if (lo == hi)
        return -1;
    if (arr[mid] > v)
        return search(arr, lo, mid, v);
    else
        return search(arr, mid, hi, v);
}
```

$T(n)$; $O(1)$; $O(1)$; $O(1)$; $T\left(\frac{n}{2}\right)$; $T\left(\frac{n}{2}\right)$

Recurenta devine:

$$T(n) = T\left(\frac{n}{2}\right) + \underbrace{\theta(1)}_{= c}$$

Metoda 1 – Teorema Master

$\Rightarrow a = 1;\ b = 2$

$\Rightarrow \log_b a = \log_2 1 = 0$

$f(n) = \theta(1) = \theta(n^0 \cdot log^0 n)$

$\Rightarrow c = 0;\ k = 0$

$\Rightarrow Master\ II \Rightarrow c = \log_2 1$

$\Rightarrow T(n) = \theta\left(n^{\log_b a} \cdot log^{k+1} n\right) = \theta(n^0 \cdot log^1 n) = \theta(log\, n)$

Metoda 2 - substitutie

Presupunem $\boldsymbol{T\left(\frac{n}{2}\right) \in \theta\left(\log \frac{n}{2}\right)}$ si aratam ca $\boldsymbol{T(n) \in \theta(\log n)}$.

$\exists\, c_1, c_2 > 0, \qquad c_1 \log\frac{n}{2} \le T\left(\frac{n}{2}\right) \le c_2 \log\frac{n}{2} \qquad | + c$

$c_1 \log\frac{n}{2} + c \le T\left(\frac{n}{2}\right) + c \le c_2 \log\frac{n}{2} + c$

$c_1 \log n \le c + c_1 \log n - c_1 \log 2 \le T(n) \le c_2 \log n - c_2 \log 2 + c \le c_2 \log n$

Putem scrie aceste inegalitati, daca relatiile obtinute intre c, c_1, c_2 nu sunt contradictorii.

$c - c_1 \log 2 \ge 0\ ;\ c - c_2 \log 2 \le 0 \ \Rightarrow\ c_1 \log 2 \le c \le c_2 \log 2$

$\Rightarrow\ \boldsymbol{T(n) \in \theta(\log n)}$

Metoda 3 – arbori

		Cost
niv 0	$T(n)$	……c
niv 1	$T\left(\frac{n}{2}\right)$	……c
niv 2	$T\left(\frac{n}{2^2}\right)$	……c
…..	...	……
niv h	$T\left(\frac{n}{2^h}\right)$	

Frunza; input de marime foarte mica

Nod = problema ce inca se poate separa in subprobleme
Frunza = subproblema ce nu se mai poate separa in subprobleme

Pe nivelul h avem frunze T(1), adica vectori cu 1 element:

$\frac{n}{2^h} = 1 \quad \Rightarrow \quad h = \log n$

Suma costurilor = Suma costurilor pe nivelele unde exista noduri + numar de frunze * costul unei frunze

la frunze, costul este constant

$\sum_{i=0}^{h-1} c + 1 \cdot T(1) = \sum_{0}^{\log n} c + T(1) =$

Nivelele unde nodurile se separa in subprobleme

numarul de frunze

$= \boldsymbol{c \cdot \log n + \theta(1) = \theta(\log n)}$

Exercitiul 4

$\boldsymbol{T(n) = 2T\left(\frac{n}{2}\right) + 1}$ **& Metoda substitutiei**

Cum ne dam seama ce complexitate sa folosim in metoda substitutiei?
Aplicam Teorema Master daca se poate:

$a = 2; b = 2;\ \log_b a = 1;\ \ f(n) = \theta(n^0) \quad \Rightarrow\ c = 0 < \log_b a$

$\Rightarrow Master\ I \qquad \boldsymbol{T(n) = \theta(n)}$

Presupunem $T\left(\frac{n}{2}\right) = \theta\left(\log\frac{n}{2}\right) \ \Rightarrow\ \exists\, c_1, c_2 > 0$, astfel incat $c_1 \cdot \frac{n}{2} \le T\left(\frac{n}{2}\right) \le c_2 \cdot \frac{n}{2}$

$\Rightarrow \quad 2 \cdot c_1 \cdot \frac{n}{2} \le T\left(\frac{n}{2}\right) \cdot 2 \le 2 \cdot c_2 \cdot \frac{n}{2} \quad |+1$

$c_1 \cdot n + 1 \le T\left(\frac{n}{2}\right) \cdot 2 + 1 \le c_2 \cdot n + 1$

$c_1 \cdot n \le \qquad\qquad \le n \cdot (c_2 + 1)$

$\Rightarrow \quad c_1 \cdot n \le T(n) \le c_2' \cdot n \qquad \forall\, n \ge 1$

$\Rightarrow\ \boldsymbol{T(n) \in \theta(n)}$

Exercitiul 5

$$T(n) = T\left(\frac{n}{4}\right) + T\left(\frac{3n}{4}\right) + n$$

NU putem aplica Teorema Master.
Desenand arborele, observam ca o ramura va ajunge la frunze mai rapid decat restul.

h_1 = primul nivel unde exista o frunza

h_2 = ultimul nivel unde exista o frunza

Vom gasi 2 margini ale lui T(n):

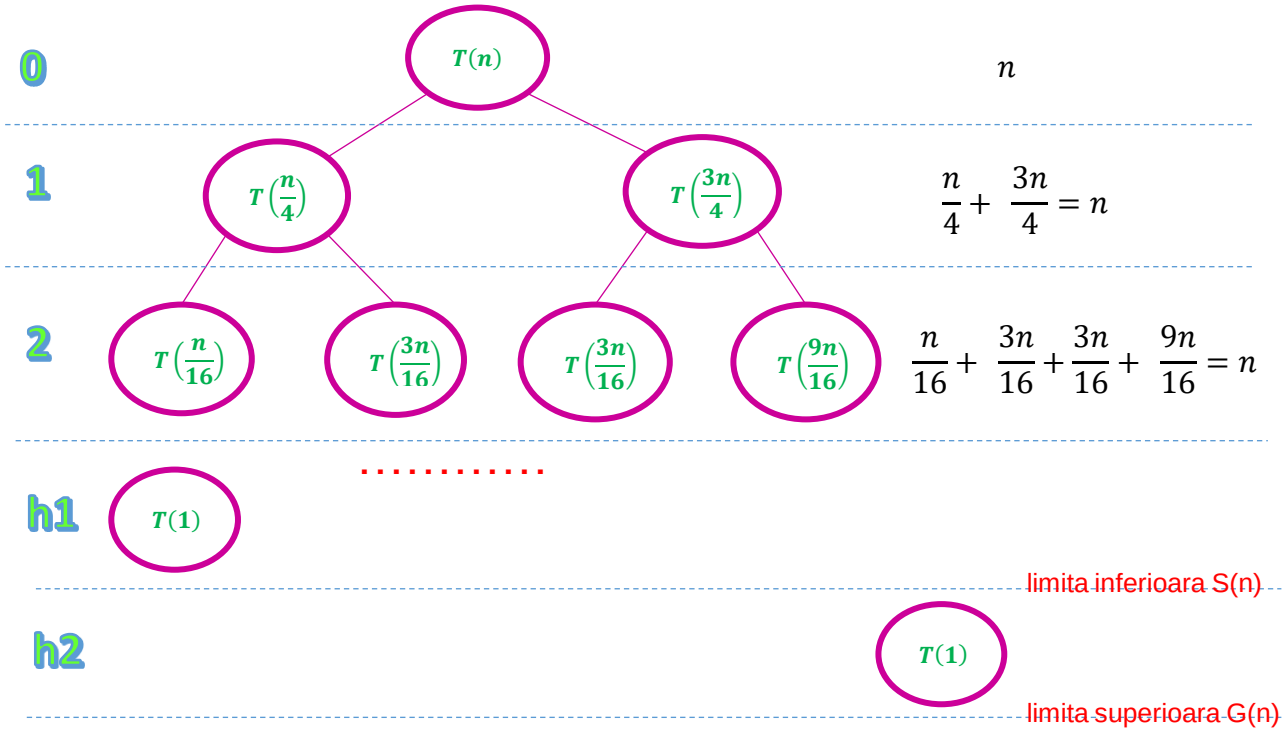
S(n): aproximare prin lipsa a lui T(n); presupunem ca nivelul h_1 este plin cu frunze

G(n): aproximare prin adaos a lui T(n); presupunem ca nivelul h_2-1 este plin cu noduri, iar nivelul h_2 are o singura frunza

$$S(n) \le T(n) \le G(n)$$

$$T(1) = T\left(\frac{n}{4}h_1\right) \quad \Rightarrow \quad h_1 = log_4\, n$$

$$T(1) = T\left(\left(\frac{3}{4}\right)^{h_2} \cdot n\right) \quad \Rightarrow \quad h_2 = \log_{\frac{4}{3}} n$$



$$S(n) = \sum_0^{h_1-1} n + T(1) \cdot 2^{h_1} = h_1 \cdot n + \theta(1) \cdot 2^{h_1} = n \cdot \log_4 n + \theta(1) \cdot 2^{\log_4 n}$$

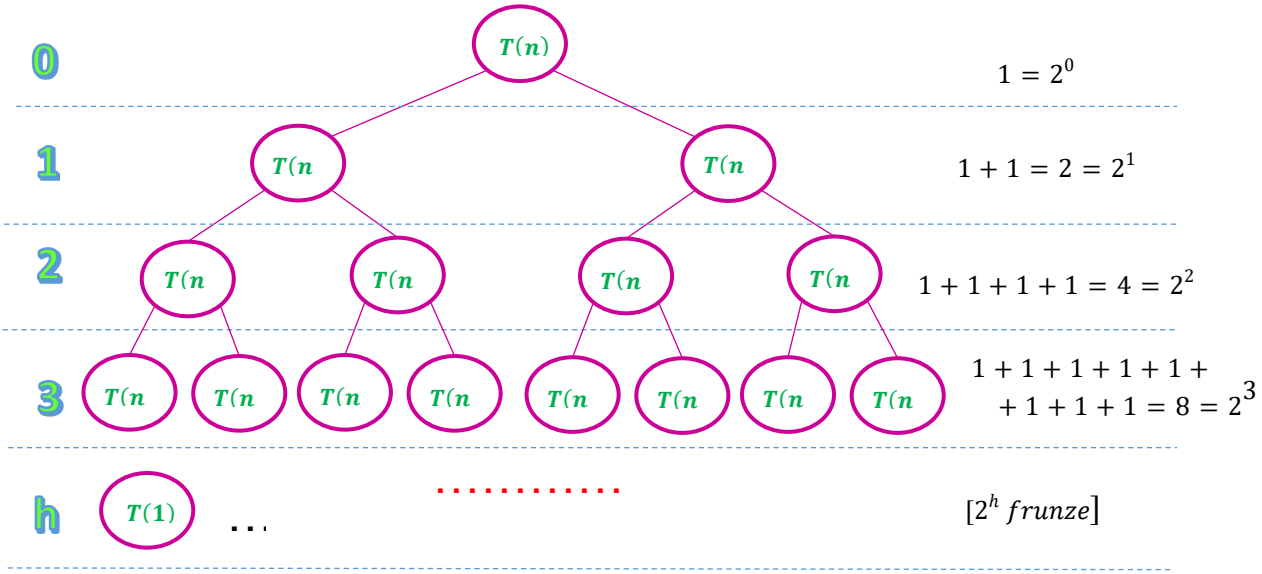
$$= n \cdot \log_4 n + \theta(1) \cdot n^{\log_4 2} = \theta(n \cdot \log n)$$

$$D(n) = \sum_0^{h_2-1} n + T(1) \cdot 1 = n \cdot h_2 + \theta(1) = n \cdot \log_{\frac{4}{3}} n = \theta(n \cdot \log n)$$

$$\Rightarrow \quad T(n) \in \theta(n \cdot \log n)$$

Exercitiul 3.1 $T(n) = 2T(n-1) + 1$

Nu putem aplica Master.



$n - h = 1 \Rightarrow h = n - 1 \Rightarrow$ total n nivele

$$C = \sum_{i=0}^{h-1} 2^i + 2^h \cdot T(1) = \frac{2^h - 1}{2-1} + 2^h \cdot O(1) = 2^h + 2^h \cdot O(1) - 1 \in \theta(2^h)$$

$$\Rightarrow T(n) \in \theta(2^h) = \theta(2^{n-1}) = \theta(2^n)$$

Exercitiul 3.2 $T(n) = T(n-1) + 1$

Nu putem aplica Master.

0: $T(n)$ 1
1: $T(n-1)$ 1
2: $T(n-2)$ 1
...
h: $T(1)$ ..... [1 frunza]

$n - 1 = h$

$$C = \sum_{i=0}^{(n-1)-1} 1 + 1 \cdot T(1) = (n-1) \cdot 1 + 1 \cdot O(1) \in \theta(n)$$

Exercitiul 3.3 $T(n) = 2T\left(\frac{n}{2}\right) + \log n$

Presupunem $T\left(\frac{n}{2}\right) \in \theta\left(\frac{n}{2}\right)$

$\exists\, c_1, c_2 > 0$ a.i. $c_1 \cdot \frac{n}{2} \le T\left(\frac{n}{2}\right) \le c_2 \cdot \frac{n}{2}$

$\Rightarrow \quad c_1 \cdot n \le 2 \cdot T\left(\frac{n}{2}\right) \le c_2 \cdot n \quad | + \log n$

$$c_1 \cdot n + \log n \le 2 \cdot T\left(\frac{n}{2}\right) + \log n \le c_2 \cdot n + \log n$$

$c_1 \cdot n \le$ $\le n \cdot (c_2 + 1)$

$de\ la\ un\ n > n_0$

$\Rightarrow$ ***Nu se obtin contradictii, deci e OK***

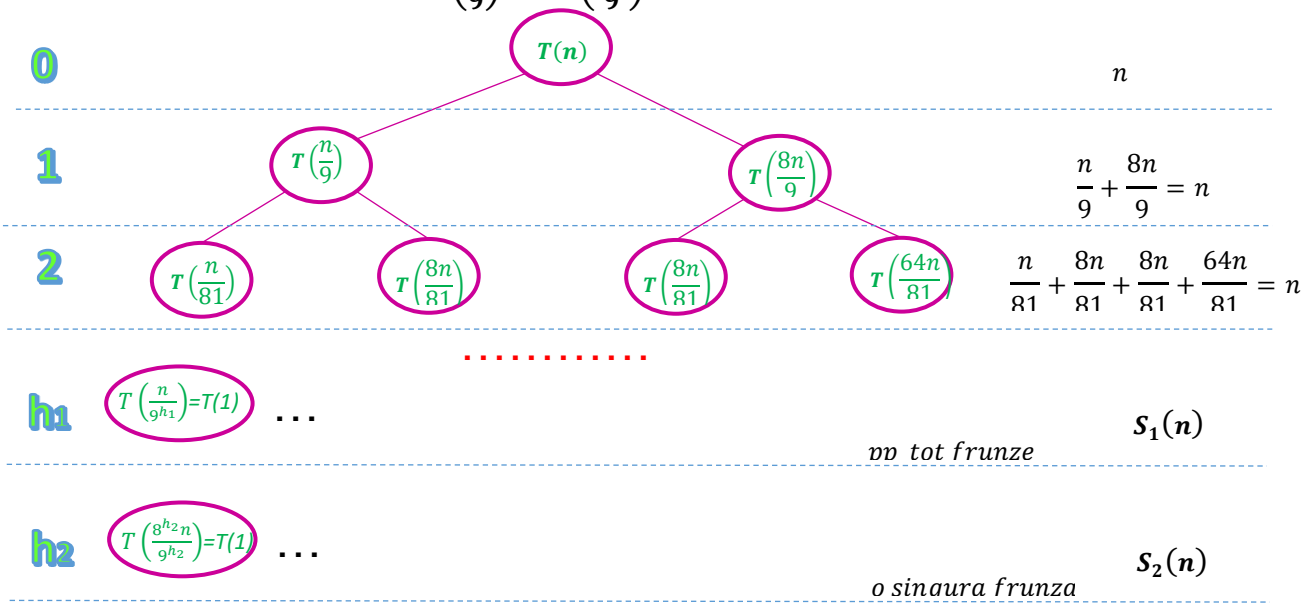
Exercitiul 3.4 $\boldsymbol{T(n) = T\left(\frac{2n}{3}\right) + 1}$

$a = 1;\ b = \frac{3}{2};\ \log_{\frac{3}{2}} 1 = 0;$

$1 = f(n) = \theta(n^0 \cdot log^0 n) \Rightarrow c = 0;\ k = 0$

$\Rightarrow Master\ II \Rightarrow \boldsymbol{T(n) \in \theta\left(n^{log_{\frac{3}{2}} 1} \cdot log^{0+1} n\right) = \theta(log\, n)}$

Exercitiul 3.5 $\boldsymbol{T(n) = T\left(\frac{n}{9}\right) + T\left(\frac{8n}{9}\right) + n}$



$\frac{n}{9^{h_1}} = 1 \Rightarrow h_1 = log_9 n$

$\frac{8^{h_2} n}{9^{h_2}} = 1 \Rightarrow h_2 = log_{\frac{9}{8}} n$

$S_1(n) \le T(n) \le S_2(n)$

$S_1(n) = \sum_0^{h_1 - 1} n + T(1) \cdot 2^{h_1} = n \cdot h_1 + 2^{h_1} \cdot T(1) = n\, log_9 n + 2^{log_9 n} \cdot O(1) \in \theta(n \log n)$

$S_2(n) = \sum_0^{h_2 - 1} n + T(1) \cdot 1 = n \cdot h_2 + O(1) = n\, log_{\frac{9}{8}} n + O(1) \in \theta(n \log n)$

$\Rightarrow \boldsymbol{T(n) \in \theta(n\, log\, n)}$

Exercitiul 3.6. $\boldsymbol{T(n) = T\left(\sqrt{n}\right) + 1}$

Expresia nu ne permite sa aplicam Master.

Notam $n = 2^x \Rightarrow x = log_2 n \Rightarrow T(2^x) = T\left(2^{x/2}\right) + 1$

Notam S(k) $= T(2^x) \Rightarrow S(x) = S\left(\frac{x}{2}\right) + 1 \Rightarrow a = 1;\ b = 2;\ c = 0;\ k = 0;\ Master\ II$

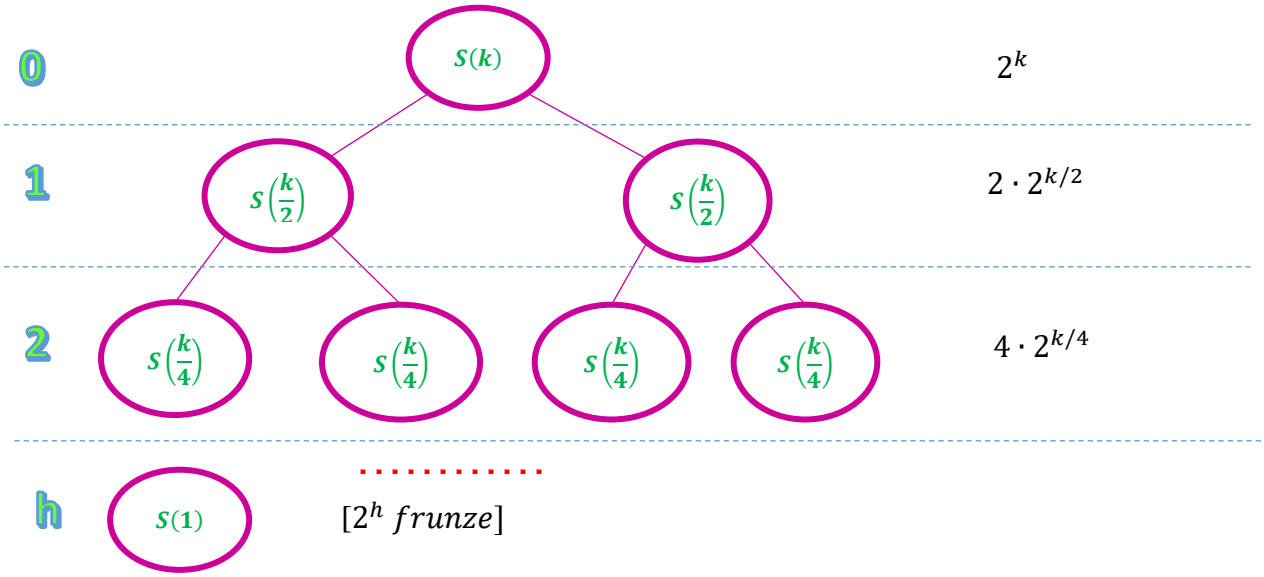
$\Rightarrow S(x) \in \theta(log\, x) \Rightarrow T(2^x) \in \theta(log\, x) \Rightarrow \boldsymbol{T(n) \in \theta(log\, x) = \theta(log\, log\, n)}$

Exercitiul 3.7. $T(n) = 2T(\sqrt{n}) + n$

Aceeasi idee de la 3.6.

Notam $n = 2^k \quad \Rightarrow \quad k = log_2 n \quad \Rightarrow \quad T(2^k) = 2T(2^{k/2}) + 2^k$

Notam S(k) $= T(2^k) \quad \Rightarrow \quad S(k) = 2S\left(\frac{k}{2}\right) + 2^k$



$\frac{k}{2^h} = 1 \quad \Rightarrow \quad h = log_2 k$

$$S(k) = \sum_{i=0}^{h-1} 2^i * 2^{\frac{k}{2^i}} + T(1) \cdot 2^h = \ldots$$

Exercitiul 3.8. $T(n) = \sqrt{n}\ T(\sqrt{n}) + n$

Notam $n = 2^k \quad \Rightarrow \quad k = log_2 n \quad \Rightarrow \quad T(2^k) = 2^{k/2}\, T(2^{k/2}) + 2^k \quad | :2^k$

$$\frac{T(2^k)}{2^k} = \frac{T(2^{k/2})}{2^{k/2}} + 1$$

Notam S(k) = $\frac{T(2^k)}{2^k} \quad \Rightarrow \quad S(k) = S\left(\frac{k}{2}\right) + 1$

$\Rightarrow \ a = 1;\ b = 2;\ c = 0;\ k = 0;\ Master\ II \Rightarrow S(k) = \theta(log\, k)$

$T(2^k) = 2^k \cdot \theta(log\, k) = \theta(2^k \cdot log\, k) \Rightarrow \boldsymbol{T(n) \in \theta(n \cdot log\, log\, n)}$